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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT		
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Abstract

The Sales/Demand Forecasting System is an intelligent data analytics and machine learning-based solution developed to forecast future sales and demand by analyzing historical sales data. In modern business environments, organizations generate large volumes of sales information on a daily basis, making it important to analyze this data effectively and identify patterns that can support better decision-making. Accurate sales forecasting enables businesses to anticipate customer demand, manage inventory efficiently, optimize resources, and improve their overall operational performance.

The proposed system focuses on analyzing historical sales records and extracting meaningful patterns such as trends, seasonality, fluctuations, and demand variations over time. The system begins with data collection and preprocessing, where missing values, inconsistencies, and irrelevant information are handled to prepare the dataset for analysis. Exploratory Data Analysis (EDA) is then performed using statistical techniques and visualizations to understand the behavior of sales over different time periods. These insights help identify recurring patterns and important factors that may influence future demand.

For forecasting, time-series analysis techniques such as ARIMA and Prophet are applied to historical sales data to predict future sales or demand. The models analyze previous observations and use identified temporal patterns to generate forecasts for upcoming periods. The forecasting models can be evaluated using suitable performance measures to compare their accuracy and determine the most effective approach for the given dataset.

The system also includes a visualization and dashboard component that presents historical sales trends, demand patterns, forecasted values, and model performance in an easy-to-understand format. Interactive charts and graphs allow users to explore the data and compare actual sales with predicted values. This makes the system useful not only for forecasting but also for understanding business trends and supporting data-driven decisions.

The proposed system aims to help businesses reduce overstocking and understocking, improve inventory planning, anticipate future demand, optimize resource allocation, and support strategic decision-making. By combining data preprocessing, exploratory analysis, time-series forecasting, model evaluation, and interactive

visualization, the Sales/Demand Forecasting System provides a complete analytical framework for understanding historical sales behavior and predicting future demand.

Signature of the Guide

HOD